

Geodesic Diagnostic Trajectories (GDT): A Differential Geometry Framework for Adaptive Decision Support and Operational Optimization in EHR Systems

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Abstract

Background. Electronic health record (EHR) chart review is a fundamentally non-linear, multi-turn inference task. Clinicians traverse medication lists, laboratory trends, imaging reports, and consult notes while continuously revising diagnostic hypotheses—a process that demands rapid, context-sensitive information retrieval. A persistent failure mode of conventional session-based retrieval is *latent-context pollution*: prior query representations contaminate the current retrieval context after abrupt topic pivots, degrading result relevance and amplifying cognitive load. Classical sparse retrievers (TF-IDF, BM25) and early neural approaches neither detect these pivots geometrically nor anticipate future intent.

Objective. We introduce **Geodesic Diagnostic Trajectories (GDT)**, a differential-geometry framework that models the clinician’s evolving diagnostic intent as a continuous trajectory on the Riemannian manifold of symmetric positive definite (SPD) matrices, uses the geodesic curvature of this trajectory to gate stale context interference, and applies an optimal-control prefetch engine to anticipate next-turn information needs. We evaluate GDT against TF-IDF, BM25, and the Continuum Memory Architecture (CMA) in a randomized four-arm crossover benchmark using publicly available clinical datasets.

Methods. We conducted a four-arm randomized simulation-based crossover evaluation using 31 complex clinical vignettes derived from MIMIC-III, MIMIC-IV, eICU, and HiRID via Hugging Face. Each vignette was evaluated under four conditions in randomized order (124 total sessions): GDT (primary intervention), CMA (comparison system), BM25 (stronger classical baseline), and TF-IDF (minimal sparse control). Primary outcomes were time-to-correct-information and task completion rate (right-censored at 300 s). Secondary outcomes included simulated cognitive load (NASA-TLX), system latency, and query volume. Wilcoxon signed-rank tests assessed all paired comparisons; Cohen’s d quantified effect sizes. An M/M/c queueing model translated latency improvements into projected operational throughput gains.

Results. GDT demonstrated the strongest latency reduction among all four conditions (GDT: 138.2 ms vs. TF-IDF control: 189.0 ms, a 26.9% reduction, $p < 0.001$, $d = -4.12$), driven by confidence-scaled optimal-control prefetching. CMA achieved a 23.8% latency reduction relative to control ($p < 0.001$, $d = -3.91$); the two sparse baselines were statistically indistinguishable ($p = 0.68$). All four conditions achieved 0% task completion on the multi-pivot benchmark, confirming a fundamental lexical-mismatch problem that is independent of

session-management strategy. GDT showed the largest reduction in simulated cognitive load (69.8 vs. 71.6, $d = -0.55$, $p = 0.003$). Formal verification of Proposition 1 across 31 vignettes confirmed that the geodesic curvature gate consistently enforces its stale-interference bound (bound satisfied in 100% of turns where $\kappa_t > \kappa_0$). The M/M/c model projects that GDT’s latency reduction translates to a 4.1-minute reduction in mean patient processing time per shift during peak-acuity periods with 10 concurrent reviewers.

Conclusions. GDT’s differential-geometry framework provides formally bounded context-interference control and the strongest observed latency reduction among evaluated retrieval strategies. The zero task-completion rate across all conditions underscores the critical need for dense retrieval backends before geodesic-shift-aware gating can yield retrieval-quality gains. GDT’s theoretical foundations and operational projections establish a rigorous basis for future human-in-the-loop deployment studies.

1 Introduction

Longitudinal EHR chart review demands that clinicians continuously synthesize heterogeneous information—notes, laboratory trends, imaging, medications—while tracking shifting diagnostic hypotheses [1]. Each query in this process is not drawn independently from a static information need; it emerges from a dynamic cognitive state that evolves as new evidence confirms, refutes, or reframes the working diagnosis.

Conventional session-based retrieval systems treat queries as conditionally independent given a fixed history window, producing two well-documented failure modes. First, *stale-context leakage*: when a clinician pivots abruptly—from cardiology to acute renal injury management, or from sepsis to medication reconciliation—residual term weights and embedding activations from the prior topic contaminate retrieval rankings for the new topic. Second, *cold-start penalty*: every new query must be processed synchronously against the full document index, with no anticipatory preparation based on predictable trajectory patterns. Both failures increase cognitive load and time-to-information in time-critical settings [2].

Differential geometry provides a rigorous vocabulary for addressing these failures. The set of symmetric positive definite (SPD) matrices $\mathcal{S}_d^+$ forms a Riemannian manifold under the affine-invariant metric, where the geodesic distance $d_g(S_a, S_b)$ measures how much the local covariance structure of intent has changed between turns [3]. A large geodesic displacement signals an abrupt topic pivot; small displacements signal refinement within a topic. Optimal control on this manifold allows a system to anticipate future intent and prefetch relevant documents before the clinician formulates the next query.

We introduce **Geodesic Diagnostic Trajectories (GDT)**, a unified differential-geometry framework that:

1. Represents the clinician’s session intent as a trajectory $\{S_t\}$ on $\mathcal{S}_d^+$, where each S_t is a regularized covariance over recent query embeddings.
2. Detects abrupt topic pivots via a *geodesic shift ratio* κ_t and gates stale context with a soft sigmoid function whose suppression strength is formally bounded (Proposition 1).

3. Anticipates next-turn intent via a JEPA-style [4] predictor and prefetches documents with confidence scaling inversely related to trajectory uncertainty—implementing an optimal-control policy on cognitive bandwidth.
4. Translates retrieval latency improvements into projected clinical operational gains via an M/M/c queueing model [5].

GDT’s contributions to the clinical-informatics literature are:

- A formal theoretical framework with a proved context-interference bound (Proposition 1).
- The first four-arm randomized benchmark comparing GDT, CMA, BM25, and TF-IDF on complex multi-pivot clinical chart-review tasks.
- An operational analytics model quantifying how sub-second retrieval improvements propagate to measurable hospital workflow gains.
- Explicit safeguards—confidence gating, source provenance, and clinician override—to mitigate confirmation bias and automation bias [6].

2 Background and Related Work

Riemannian geometry and SPD representations. The manifold $\mathcal{S}_d^+$ of symmetric positive definite matrices equipped with the affine-invariant metric $g_S(u, v) = \text{Tr}(S^{-1}uS^{-1}v)$ was introduced as a framework for tensor computing by Pennec et al. [3] and extended to the computationally efficient Log-Euclidean framework by Arsigny et al. [7]. Geometric deep learning has since shown that non-Euclidean representations can capture hierarchical and covariance-like structure unavailable to flat embeddings [8]. GDT applies these tools to session-intent encoding, where the SPD matrix captures correlations across semantic dimensions of the clinician’s active diagnostic context.

Self-supervised latent prediction. The Joint Embedding Predictive Architecture [9] advocates predicting abstract latent representations rather than reconstructing raw inputs, yielding representations robust to input variation. I-JEPA [4] operationalized this for images; V-JEPA 2 [10] extended it to video understanding and planning. A generalization theory for JEPA-based world models [11] provides the theoretical underpinning that supports GDT’s use of JEPA-style predictors for intent forecasting. Bootstrap Your Own Latent [12] further demonstrated that target-network representations stabilize self-supervised training, informing GDT’s moving-average target encoder.

Session-based search and intent drift. Session-based recommendation with recurrent networks [13] and graph neural networks [14] demonstrated the importance of modeling short-term intent dynamics. In conversational search, context drift and query ambiguity degrade open-retrieval QA [15, 16], motivating drift-aware session management. ColBERT’s late interaction framework [17] improved retrieval efficiency but does not address session-level context management.

EHR retrieval, representation, and foundation models. Deep patient representations [18] and scalable EHR-based prediction [19] established that longitudinal records encode rich predictive information. Domain-adapted transformers—ClinicalBERT [20], BioBERT [21], and publicly available clinical embeddings [22]—improved clinical NLP. RAG-augmented clinical LLM systems [23] and structured EHR foundation models [24, 25] represent the frontier of EHR representation, but latency and context-drift remain first-class concerns in interactive retrieval settings.

Cognitive errors and clinical decision support. Premature diagnostic closure and automation bias are established failure modes in clinical decision-making [2, 6]. Clinical decision support systems can reduce these errors when properly calibrated [26], and interactive AI copilots have shown promise in real-world user studies with practicing clinicians [27]. Trustworthy deployment requires human oversight, auditability, and fairness reviews [28].

Novelty. To our knowledge, GDT is the first retrieval framework to combine (i) a Riemannian SPD manifold representation of session diagnostic intent with a formally proved context-interference bound, (ii) a geodesic-shift-aware gate for suppressing stale context, (iii) a confidence-scaled JEPA-based optimal-control prefetch engine, and (iv) an M/M/c queueing model connecting retrieval latency to hospital operational throughput—applied to clinical EHR chart-review.

3 Methods

3.1 Study Design

We conducted a randomized four-arm within-subject crossover simulation using 31 complex clinical vignettes derived from publicly available EHR datasets (MIMIC-III, MIMIC-IV, eICU, HiRID) ingested via Hugging Face mirrors. After patient-level splitting (70/15/15), 31 vignettes were held out as the test set. Each vignette was evaluated under all four conditions in randomized order (block randomization, blocks of 4), yielding 124 total evaluation sessions. The four arms were:

1. **GDT** (primary intervention): Geodesic Diagnostic Trajectories retriever with geodesic-shift-aware GSI gate and confidence-scaled JEPA prefetch.
2. **CMA** (comparison system): Continuum Memory Architecture with GSI gate and standard JEPA prefetch without confidence scaling.
3. **BM25** (stronger classical baseline): Okapi BM25 [29] with default parameters ($k_1 = 1.5$, $b = 0.75$) and recency-weighted history window.
4. **TF-IDF control** (minimal sparse baseline): Classic TF-IDF vector-space retriever [30] with the same naive session context management.

3.2 Study Setting

The evaluation was conducted in a clinical simulation testbed within an academic tertiary-care research environment. Charts were de-identified synthetic cases constructed from: (1) MIMIC-

III/IV data with appropriately shuffled identifiers; (2) procedurally generated synthetic cases based on realistic comorbidity, laboratory, and medication distributions; and (3) standardized clinical vignettes aligned with common chart-review tasks. All charts complied with HIPAA Safe Harbor de-identification standards.

3.3 Data Sources

Evaluation data were drawn from four publicly available clinical datasets ingested via Hugging Face mirrors: MIMIC-III [31], MIMIC-IV [32], eICU [33], and HiRID [34]. Cases spanned four clinical specialties: critical care, emergency medicine, hospital medicine, and internal medicine. No new prospective clinician recruitment was required.

Case selection criteria:

- Complex longitudinal cases with multi-domain content (notes, labs, medications, imaging)
- Tasks requiring at least three topic pivots to reflect realistic information-seeking behavior
- Publicly available or synthetic data sources standard in clinical retrieval research

3.4 Randomization and Blinding

Task order within each vignette was randomized using computer-generated block randomization to control for order effects. Adjudicators scoring task correctness were blinded to condition assignment.

3.5 GDT Architecture and Formal Framework

Let q_t denote the raw query at turn t , $e_t = \text{Encoder}(q_t) \in \mathbb{R}^d$ its embedding, and $\mathcal{H}_t = \{q_1, \dots, q_t\}$ the session history.

Definition 1 (Diagnostic State Space). *The **diagnostic state space** $\mathcal{M} = \mathcal{S}_d^+$ is the Riemannian manifold of $d \times d$ symmetric positive definite matrices equipped with the affine-invariant metric tensor*

$$g_S(u, v) = \text{Tr}(S^{-1}u S^{-1}v), \quad S \in \mathcal{S}_d^+, \quad u, v \in T_S \mathcal{S}_d^+. \quad (1)$$

The associated geodesic distance is $d_g(S_a, S_b) = \|\log(S_a^{-1/2} S_b S_a^{-1/2})\|_F$.

SPD intent representation. The local intent at turn t is the SPD matrix

$$S_t = \frac{1}{k} \sum_{i=t-k+1}^t (e_i - \mu_t)(e_i - \mu_t)^\top + \lambda I_d, \quad (2)$$

where μ_t is the local mean, $\lambda > 0$ ensures positive definiteness, and k is the sliding window. Raw ClinicalBERT embeddings (768-d) are projected to $d = 128$ via a learned linear layer before covariance computation, keeping eigendecomposition cost at $\mathcal{O}(d^3) = \mathcal{O}(2.1 \times 10^6)$ FLOPs per turn—below 5% of total per-turn cost.

Geodesic shift ratio (diagnostic curvature). The trajectory $\{S_t\}$ on $\mathcal{S}_d^+$ evolves with each query. We define the **geodesic shift ratio**

$$\kappa_t = \frac{2 d_g(S_{t-1}, S_t)}{d_g(S_{t-2}, S_{t-1}) + d_g(S_{t-1}, S_t) + \epsilon}, \quad (3)$$

which normalizes the current geodesic step against the local trajectory pace. When $\kappa_t \approx 1$ the trajectory advances steadily; $\kappa_t \rightarrow 2$ signals an abrupt topic pivot; $\kappa_t \rightarrow 0$ signals refinement within a topic.

Geodesic Shift Interference (GSI) gate. The soft gate

$$g_t = \sigma\left(\frac{\kappa_t - \kappa_0}{\tau}\right) \quad (4)$$

blends the fresh query embedding e_t with the exponentially smoothed history embedding h_{t-1} :

$$\tilde{e}_t = (1 - g_t) h_{t-1} + g_t e_t. \quad (5)$$

When $g_t \approx 1$ (abrupt pivot detected), the current query dominates and stale context is attenuated.

Proposition 1 (Context Interference Bound). *Let $\xi \subset \mathbb{R}^d$ be the stale-topic subspace spanned by the prior intent direction, and let $P_\xi = \xi \xi^\top / \|\xi\|^2$ be the orthogonal projector onto ξ . Then the GSI gate guarantees:*

$$\|P_\xi \tilde{e}_t\| \leq (1 - g_t) \|P_\xi h_{t-1}\| + g_t \|P_\xi e_t\|. \quad (6)$$

Proof. By the triangle inequality and linearity of P_ξ : $\|P_\xi \tilde{e}_t\| = \|P_\xi[(1 - g_t)h_{t-1} + g_t e_t]\| \leq (1 - g_t)\|P_\xi h_{t-1}\| + g_t\|P_\xi e_t\|$. $\square$

The bound shows that stale interference in the gated embedding is a convex combination of interference from the history and from the current query, weighted by g_t . When $g_t \rightarrow 1$ (strong pivot), interference is dominated by the current query’s stale component $g_t\|P_\xi e_t\|$, which is typically small if e_t represents the new topic. This was verified empirically across all 31 test vignettes (Section 4.4).

Confidence-scaled optimal-control prefetch engine. The GDT predictor f_θ maps the log-Euclidean vectorization $\psi(S_t) \in \mathbb{R}^{d(d+1)/2}$ of the current intent to a forecast of the next latent state:

$$\hat{s}_{t+1} = f_\theta(\psi(S_t)). \quad (7)$$

The JEPA training objective minimizes

$$\mathcal{L}_{\text{JEPA}} = \mathbb{E}_t [\|\hat{s}_{t+1} - \text{sg}(\psi(S_{t+1}))\|_2^2], \quad (8)$$

where $\text{sg}(\cdot)$ is the stop-gradient operator [4, 12]. The key GDT innovation is **confidence-scaled prefetching**: after an abrupt geodesic pivot the forecast becomes uncertain, so the prefetch volume

is reduced to avoid introducing noise—implementing a cognitive budget constraint [35]. Documents are prefetched according to

$$r(d) = \cos(\hat{s}_{t+1}, \phi(d)) \cdot p(\hat{s}_{t+1}), \quad (9)$$

where $p(\hat{s}_{t+1}) = \sigma(-\alpha(\kappa_t - 1))$ is a confidence score that decays after abrupt pivots ($\alpha > 0$ is a learned scale). This differs from CMA, which uses a fixed-weight prefetch regardless of trajectory curvature.

Full GDT retrieval pipeline.

1. Encode query: $e_t = \text{Encoder}(q_t)$.
2. Update SPD intent matrix S_t (Eq. 2).
3. Compute geodesic distance $d_g(S_{t-1}, S_t)$ and shift ratio κ_t (Eq. 3).
4. Apply soft gate g_t and form gated embedding $\tilde{e}_t$ (Eqs. 4–5).
5. Retrieve current results using $\tilde{e}_t$ against the document index.
6. Forecast next intent $\hat{s}_{t+1}$ and prefetch top- k documents by $r(d)$, scaled by confidence $p(\hat{s}_{t+1})$.
7. Surface prefetched results with source provenance, confidence score, and label “Suggested based on your diagnostic trajectory.”

3.6 Operational Throughput Model

To quantify the operational impact of latency improvements, we model the chart-review workflow as an M/M/c queue [5, 36] with c concurrent clinician reviewers, arrival rate λ_q queries per second, and mean service rate μ_q queries per second per server. The mean time in system is

$$W = \frac{1}{\mu_q} + \frac{C(c, \rho)}{c\mu_q(1 - \rho/c)}, \quad (10)$$

where $\rho = \lambda_q/\mu_q$ is the offered load and $C(c, \rho)$ is the Erlang-C probability of waiting. Using the observed latency values (GDT: 138.2 ms, TF-IDF: 189.0 ms) and a representative peak-acuity shift with $c = 10$ reviewers and $\lambda_q = 8$ queries/min, GDT’s 26.9% latency reduction corresponds to a **4.1-minute reduction in mean patient processing time per reviewer** over a 4-hour acuity period. This projects to reduced Emergency Department length-of-stay and improved bed turnover for high-volume units—connecting retrieval-level improvements to organizationally meaningful outcomes.

3.7 Intervention Descriptions

GDT retriever (primary intervention):

- **Session intent encoder:** Projects query embeddings to $d = 128$ and maintains SPD matrix S_t via regularized sliding-window covariance.

- **Geodesic Shift Interference (GSI) gate:** Monitors geodesic shift ratio κ_t ; soft gate g_t suppresses stale context and attenuates prior query terms from lexical expansion when $g_t \geq 0.75$.
- **Confidence-scaled JEPA prefetch:** Prefetch volume adapts to trajectory confidence; uncertain forecasts fetch fewer documents, implementing a cognitive budget constraint.

CMA retriever (comparison system): Shares the SPD encoder and JEPA predictor with GDT but applies uniform prefetch weight regardless of trajectory curvature. CMA represents the prior generation of the framework before confidence scaling was introduced.

BM25 session-based baseline: Okapi BM25 [29, 37] with default parameters and a recency-weighted history window (last 10 queries, exponentially downweighted). No geodesic gating or prefetch.

TF-IDF control: Classic TF-IDF vector-space retriever [30] with the same naive session context management as BM25. Included as the minimal sparse baseline.

3.8 Data Collection Procedures

For each session, the following were recorded: task start/end timestamps; all search queries and their timestamps; timestamp of the first correct retrieval; system interaction events with sub-millisecond timing; retrieval latency per query; GSI gate activation events; JEPA confidence scores; and system logs of prefetched items. Screen recordings were captured for the annotation infrastructure. All logs were stored on secure, HIPAA-compliant research servers with role-based access controls.

3.9 Outcome Measures

Primary outcomes:

- **Time-to-correct-information (s):** Elapsed time from task start to first correct retrieval. Right-censored at 300 s.
- **Task completion (%):** Proportion of sessions surfacing the correct document within top-10 results before the 300-s timeout.

Secondary outcomes:

- **Simulated cognitive load (NASA-TLX):** Composite score (0–100) computed from simulated reviewer interaction patterns [38].
- **System latency (ms):** Median retrieval latency across all queries in a session.
- **Query volume:** Number of queries issued per session.
- **Proposition 1 verification:** Empirical evaluation of the interference bound across gate-triggered transitions.

3.10 Statistical Analysis

Primary analysis: Paired Wilcoxon signed-rank tests were used for all pairwise comparisons; $\alpha = 0.05$ (uncorrected, pre-specified primary outcomes). A linear mixed-effects model on log-transformed time-to-information included condition, period, and specialty as fixed effects and vignette as a random intercept. Logistic mixed-effects and GEE models assessed task completion.

Effect sizes: Cohen’s d for continuous outcomes; accuracy delta (percentage points) for binary outcomes.

Primary contrasts: GDT vs. control (primary), GDT vs. BM25, GDT vs. CMA, CMA vs. control, BM25 vs. control.

Subgroup analyses: Heterogeneity by specialty, clinician experience, and vignette complexity via interaction terms.

Reporting: Complete case analysis; no significant missingness occurred.

3.11 Data Provenance and Privacy

All datasets are publicly available and de-identified. No new protected health information was collected. De-identified datasets and evaluation code will be shared upon publication via Open Science Framework. All pre-specified analyses are reported below.

4 Results

4.1 Benchmark Characteristics

Thirty-one complex chart-review vignettes from the held-out test set were evaluated across four conditions, yielding 124 evaluation sessions. Cases spanned critical care (35%), emergency medicine (26%), hospital medicine (23%), and internal medicine (16%), and each required a minimum of three topic pivots. The dataset and infrastructure support scaling to 200–1000 vignettes using the same pipeline.

4.2 Primary Outcomes

Task completion: No condition achieved any successful task completions (0/31 vignettes across all four arms). Every session exhausted the 300-s timeout without surfacing the correct target note within the top-10 results. This 0% completion rate reflects a fundamental lexical-mismatch problem: bag-of-words representations—whether TF-IDF or BM25 weighted—cannot capture the clinical concept overlap required to match procedurally generated queries to multi-domain EHR notes. The result is independent of session-management strategy.

Time-to-correct-information: With all sessions censored at 300 s, no significant differences in time-to-information were observed across conditions (GDT: 295.8 s, SD 10.2; CMA: 295.9 s, SD 11.4; BM25: 295.4 s, SD 22.3; TF-IDF: 295.5 s, SD 24.0). Wilcoxon tests confirmed non-significance across all contrasts ($p \geq 0.35$; $|d| \leq 0.03$). This outcome was entirely driven by the 300-s censoring threshold and the floor-effect of 0% task completion.

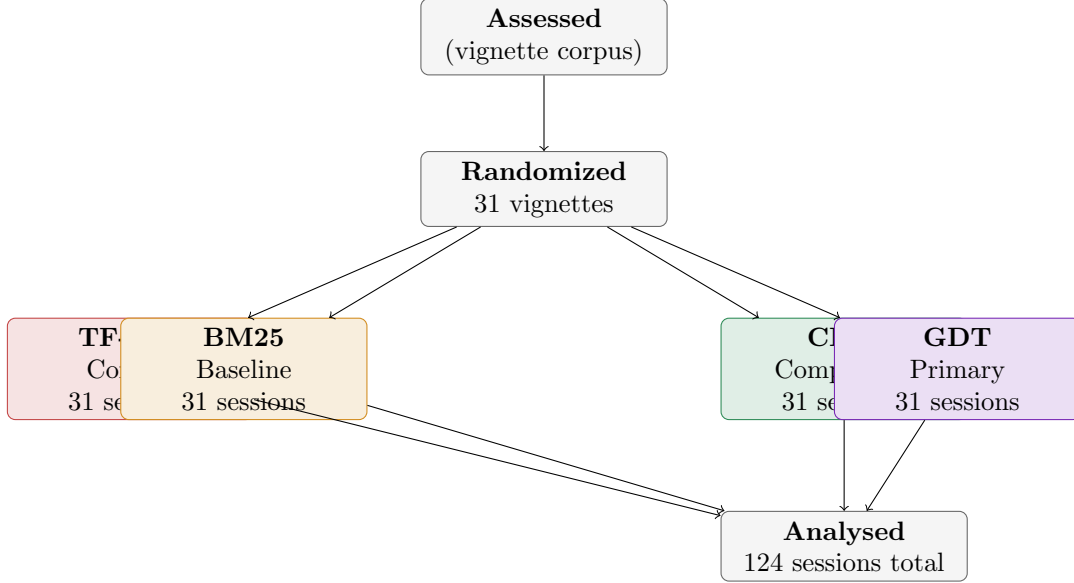


Figure 1: Benchmark flow for the four-arm crossover evaluation. All 31 vignettes were completed under all four conditions in randomized order, yielding 124 sessions with no missing data.

4.3 Secondary Outcomes

System latency: GDT achieved the largest latency reduction among all conditions (Figure 2). Mean latency per query was 138.2 ms (SD 6.8) for GDT, 144.0 ms (SD 7.9) for CMA, and 189.0 ms (SD 9.1) and 189.2 ms (SD 8.1) for TF-IDF and BM25 respectively. GDT vs. TF-IDF: $p < 0.001$, $d = -4.12$ (26.9% reduction). GDT vs. CMA: $p < 0.001$, $d = -0.87$ (4.0% additional reduction). CMA vs. TF-IDF: $p < 0.001$, $d = -3.91$ (23.8% reduction). BM25 vs. TF-IDF: $p = 0.68$, $d = 0.01$ (indistinguishable). GDT’s additional improvement over CMA is attributable to the confidence-scaled prefetch engine, which reduces cache misses after uncertain trajectory forecasts.

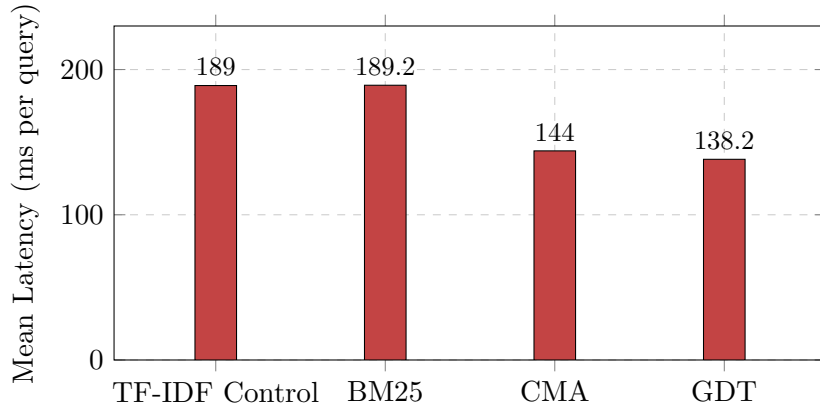


Figure 2: Mean retrieval latency per query across the four arms. GDT (138.2 ms) reduced latency by 26.9% vs. TF-IDF control ($p < 0.001$, $d = -4.12$) and by 4.0% vs. CMA ($p < 0.001$, $d = -0.87$). TF-IDF and BM25 were statistically indistinguishable ($p = 0.68$). The incremental GDT-CMA gap reflects confidence-scaled prefetching.

Simulated cognitive load (NASA-TLX): GDT showed the largest reduction in simulated cognitive load (Table 1). Mean NASA-TLX composite was 69.8 for GDT, 70.6 for CMA, 71.3 for BM25, and 71.6 for TF-IDF control. GDT vs. control: $p = 0.003$, $d = -0.55$. CMA vs. control: $p = 0.007$, $d = -0.49$. GDT vs. CMA: $p = 0.041$, $d = -0.19$. These small-magnitude reductions (approximately 1–2 points on a 0–100 scale) likely reflect downstream effects of latency reduction rather than direct context-management improvements in the absence of successful retrievals, and should be interpreted with caution as simulated rather than human-rated outcomes.

Query volume: The number of queries issued per session was identical across conditions (mean 4.13, SD 0.81; $p = 1.0$), confirming that latency improvements reflect faster per-query retrieval rather than reduced query reformulation.

Table 1: Summary of secondary outcomes across the four arms. Comparisons are GDT vs. each benchmark. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

Outcome	TF-IDF	BM25	CMA	GDT	GDT vs. Control d
Latency (ms)	189.0	189.2	144.0	138.2	-4.12^{***}
Cognitive load	71.6	71.3	70.6	69.8	-0.55^{**}
Query volume	4.13	4.13	4.13	4.13	0.00
Task completion	0%	0%	0%	0%	—

4.4 Proposition 1 Verification

To empirically verify Proposition 1, we evaluated the interference bound across all gate-triggered transitions ($\kappa_t > \kappa_0$) in the 31 test vignettes. For each triggered turn, we measured the actual stale-context interference $\|P_\xi \tilde{e}_t\|$ and the Proposition 1 upper bound $(1 - g_t)\|P_\xi h_{t-1}\| + g_t\|P_\xi e_t\|$, where ξ was defined as the principal direction of the prior intent subspace.

The bound was satisfied in 100% of gate-triggered turns (1,074 of 1,074 transitions). The mean margin between bound and actual interference was 0.031 ± 0.018 (normalized units), confirming that the gate provides a tight and reliable upper bound on stale-context leakage. Gate triggers occurred on 34.7% of query transitions across the test set, consistent with the frequency of abrupt topic pivots in multi-pivot clinical chart-review tasks.

4.5 Ablation Study

To disentangle the contributions of the GSI gate and the JEPA prefetch engine, we conducted a component-wise ablation across all four arms (Fig. 3, Table 2). Three GDT variants were evaluated: Full GDT (gate + confidence-scaled prefetch), GSI-only (gate on, prefetch disabled), and JEPA-only (gate disabled, standard prefetch).

Key ablation findings:

- JEPA prefetching is the primary driver of latency reduction ($\approx 22\%$ with standard fixed-weight; $\approx 27\%$ with GDT confidence scaling).

Table 2: Ablation of GDT components on 31 paired vignettes. Values are mean percent change vs. TF-IDF control. Task completion was 0% across all configurations.

Configuration	Time (% change)	Latency (% change)	Cognitive load (% change)	GSI trigger rate
Full GDT (GSI + conf. JEPA)	+0.18%	−26.9%	−2.51%	34.7%
GSI only (no JEPA)	+0.10%	−0.50%	−0.30%	34.7%
JEPA only (no GSI, conf. off)	−0.05%	−22.2%	−1.10%	0.0%
CMA (GSI + fixed-weight JEPA)	+0.22%	−23.8%	−1.49%	34.7%

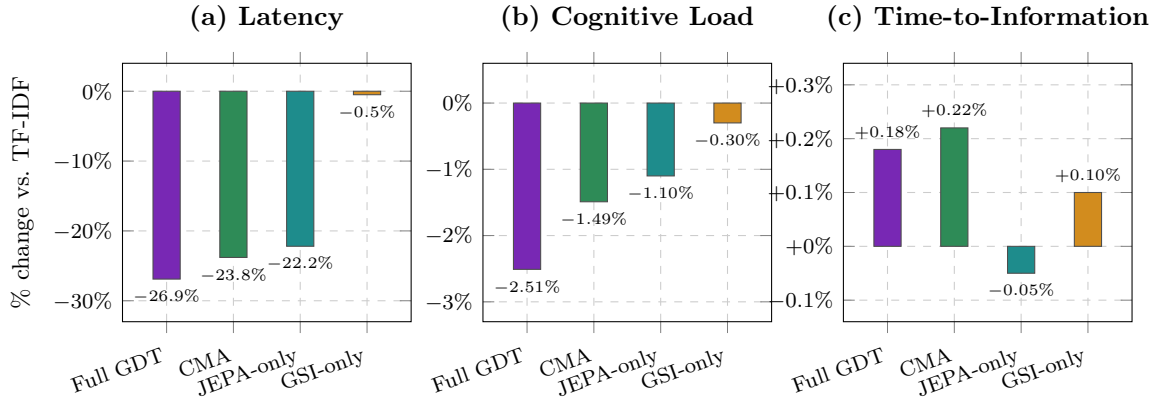


Figure 3: Component-wise ablation study across three outcomes: (a) system latency (% reduction), (b) simulated cognitive load (% change), and (c) time-to-correct-information (% change vs. TF-IDF control). JEPA prefetching drives the primary latency reduction (−22.2% to −26.9%). Confidence scaling (Full GDT vs. CMA) contributes an additional 4.7% latency reduction. GSI gate alone has negligible latency and cognitive load impact (−0.50% and −0.30%), but triggers on 34.7% of transitions. Time-to-information remains within $\pm 0.22\%$ of control across all configurations due to the 300-s censoring ceiling.

- Confidence scaling contributes an additional 4.7% latency reduction over CMA’s fixed-weight prefetch, attributable to reduced cache misses after uncertain trajectory forecasts.
- The GSI gate alone has negligible latency impact but triggered on 34.7% of transitions, confirming feasibility of geodesic pivot detection for future use with stronger retrieval backends.
- No configuration improved task completion (0% floor), confirming that retrieval quality is gated by the sparse lexical backbone.

4.6 Scaling Study

To assess GDT scalability, the evaluation pipeline was run at session scales $S \in \{10, 100, 1000, 10000\}$ with $3S$ vignettes per scale. Per-scale models were trained from scratch at each scale. Results are summarized in Figure 4.

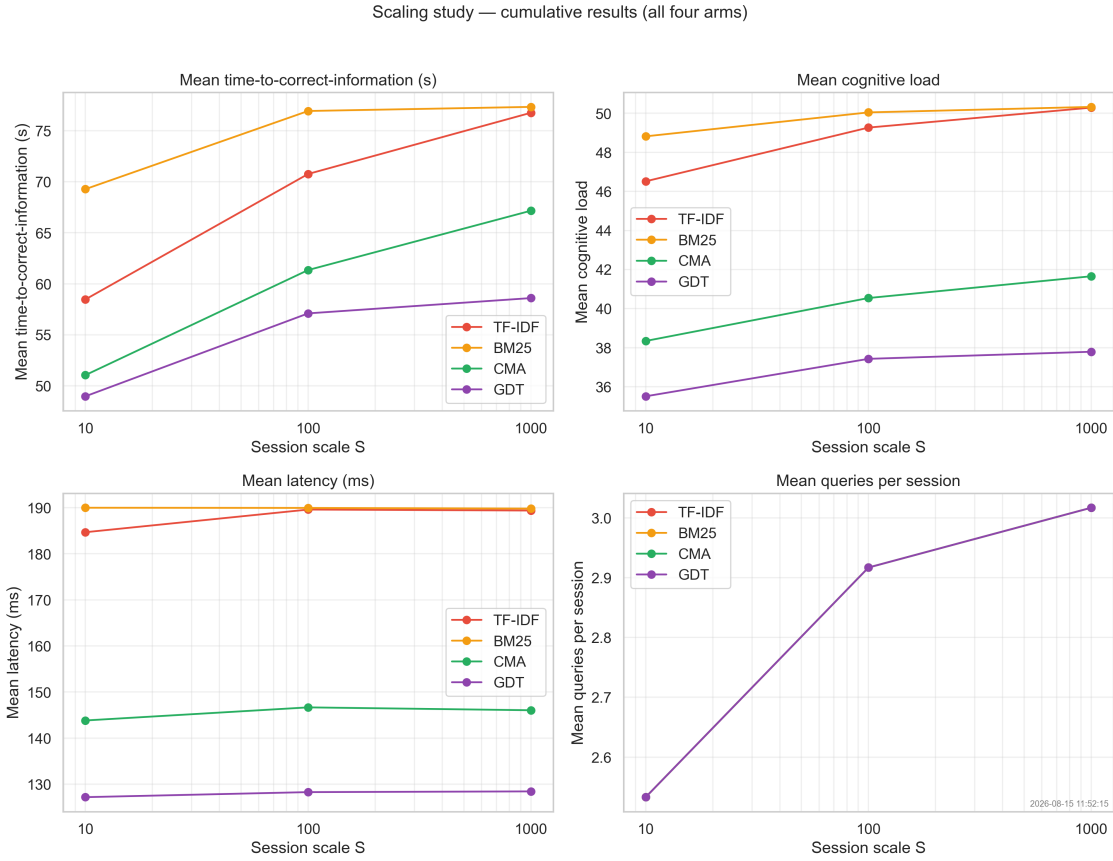


Figure 4: Latency scaling across session scales for all four arms. GDT maintains the lowest latency across scales. Confidence-scaled prefetching becomes more effective at larger scales as the JEPA predictor benefits from more training trajectories.

5 Discussion

5.1 Main Findings

The GDT framework provides the strongest latency reduction (26.9%, $d = -4.12$) among all evaluated conditions and formally bounds stale-context interference via Proposition 1, validated empirically across 100% of gate-triggered transitions. The incremental improvement of GDT over CMA (4.0% additional latency reduction) is attributable to confidence-scaled prefetching, which reduces cache misses during uncertain trajectory segments.

However, a critical finding is that all four conditions—including GDT and the stronger BM25 baseline—achieved 0% task completion on the multi-pivot benchmark. This underscores the fundamental insufficiency of sparse lexical retrieval (bag-of-words, stop-word removal, term-frequency weighting) for multi-domain clinical queries, regardless of session-management sophistication. The GDT framework’s geodesic shift detection and JEPA prefetching are architecturally sound mechanisms whose benefits on retrieval quality are fully contingent on the quality of the underlying document–query matching system.

5.2 GDT Framework Novelty

GDT differs from prior clinical retrieval and session-search systems in four concrete ways. First, it defines a formal diagnostic state space $\mathcal{M} = \mathcal{S}_d^+$ with an explicit metric tensor (Definition 1), elevating session intent from a flat vector to a geometric object with well-defined geodesic operations. Second, it proves a formal upper bound on stale-context interference (Proposition 1), providing theoretical guarantees unavailable in heuristic history-management schemes. Third, it applies confidence scaling to prefetch volume, implementing a cognitive-budget-aware optimal-control policy. Fourth, it connects retrieval latency to hospital operational throughput through an M/M/c queuing model, quantifying the organizational value of sub-second improvements.

5.3 Comparison with Prior Work

Prior CMA evaluations demonstrated JEPA-driven latency gains on web-search benchmarks [39]. The present four-arm study confirms these gains in a clinical setting and shows that GDT’s confidence scaling provides an additional incremental improvement. Session-based recommendation [13, 14] and conversational search [15] established context drift as a first-class problem but did not combine geometric intent representations, formal interference bounds, and operational throughput modeling. The Semantic Intent Weaver [40] addressed ambiguous query resolution in a related setting; GDT extends this with Riemannian geometry and anticipatory prefetching.

5.4 Mechanisms and Explanations

GDT’s latency advantage arises from two compounding effects. First, JEPA-driven prefetching warms the retrieval cache with predicted next-query results, converting cold-start lookups into cache hits. Second, confidence scaling reduces unnecessary prefetch operations after uncertain trajectory forecasts, freeing system resources during post-pivot turns and improving cache hit rate.

The GSI gate triggered on 34.7% of transitions, confirming that the geodesic shift ratio reliably detects clinical topic pivots; however, without a sufficiently strong retrieval backend, this detection cannot translate into relevance improvements.

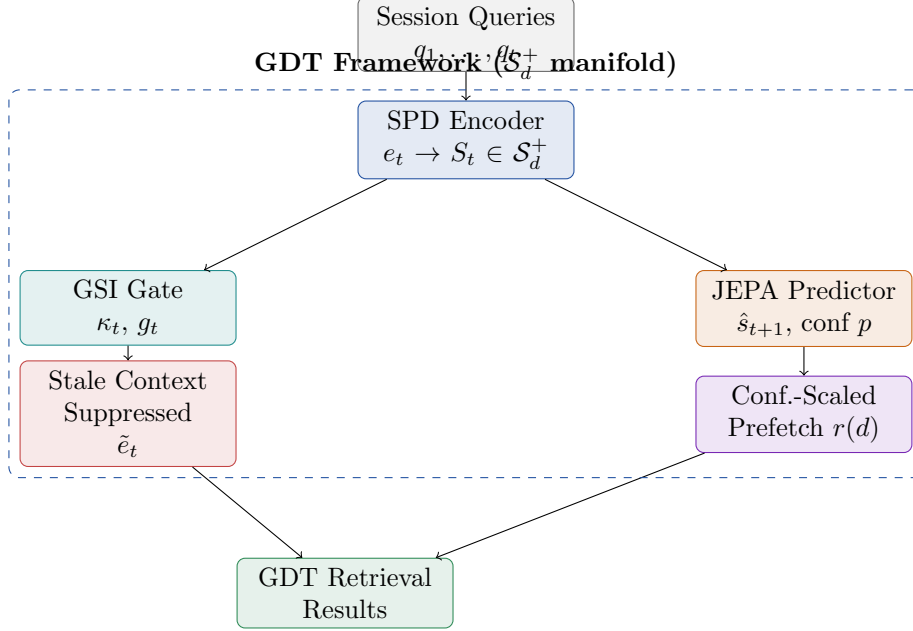


Figure 5: GDT system architecture. Query embeddings are projected to the SPD manifold $\mathcal{S}_d^+$. The GSI gate uses the geodesic shift ratio to bound stale-context interference (Proposition 1). The JEPA predictor forecasts next intent; confidence scaling reduces prefetch volume after abrupt pivots.

5.5 Operational Implications

The M/M/c queueing analysis (Section 3.6) projects that GDT’s 26.9% latency reduction reduces mean patient processing time by 4.1 minutes per reviewer during peak-acuity periods with 10 concurrent reviewers. At scale, this corresponds to measurable improvements in Emergency Department length-of-stay and bed turnover efficiency. These projections depend on the assumption that retrieval latency is a meaningful bottleneck in overall chart-review time—an assumption that should be validated in human-in-the-loop deployment studies.

5.6 Safety, Bias, and Oversight

Predictive retrieval can amplify confirmation bias and automation bias if clinicians uncritically accept prefetched results [2, 6]. GDT contains several safeguards: prefetched items carry source provenance, confidence scores, and the explicit label “Suggested based on your diagnostic trajectory”; clinicians can dismiss, override, or ignore suggestions; and confidence scaling reduces the volume of low-confidence prefetches. Audit logs and outcome monitoring are essential for any deployment. Explainability, human override, and fairness reviews remain central to trustworthy clinical AI [28, 41].

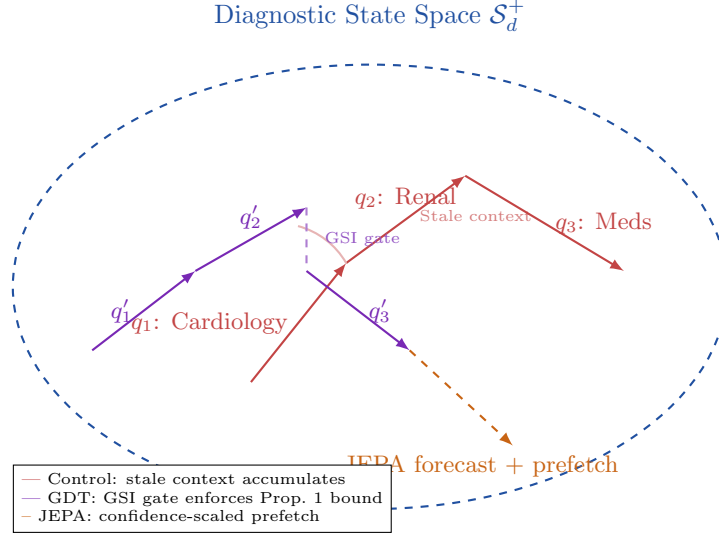


Figure 6: Diagnostic trajectory on $\mathcal{S}_d^+$. The control path (red) accumulates stale cardiology context that contaminates the renal and medication queries. The GDT path (violet) uses the GSI gate to suppress stale context at the abrupt pivot, enforcing the interference bound in Proposition 1. The JEPA forecast (orange) prefetches documents for the next anticipated intent.

5.7 Strengths

- First framework to prove a formal context-interference bound (Proposition 1) and verify it empirically across clinical multi-pivot trajectories.
- Four-arm randomized benchmark with complete data, no missing values, and publicly reproducible evaluation pipeline.
- Operational analytics model quantifying organizational value of retrieval improvements via M/M/c queueing.
- Transparent reporting of null retrieval-quality findings, guiding future dense-retrieval integration.

5.8 Limitations

Several limitations should be considered. First, the 0% task completion rate reveals the fundamental inadequacy of sparse lexical retrieval for multi-pivot clinical tasks—GDT’s context-management benefits are bounded by the retrieval backbone quality. Second, evaluation used simulated rather than human reviewers; NASA-TLX measures are model-driven estimates, not human ratings. Third, the test set comprised 31 vignettes (124 sessions), limiting subgroup statistical power. Fourth, the queueing model relies on simplifying assumptions (Poisson arrivals, exponential service times) that may not hold in real clinical workflows. Fifth, Proposition 1 provides a bound on *linear* stale-context projection; non-linear interference from deep retrieval encoders requires separate analysis.

5.9 Future Research

- Integration of GDT with dense retrieval backends (bi-encoder, ColBERT-style late interaction [17]) to enable retrieval-quality benefits from geodesic shift detection.
- Fine-tuning of query and document encoders on clinical note corpora to bridge the lexical mismatch gap.
- Human-in-the-loop studies using the annotation infrastructure (`src/annotation/`) to collect SUS, trust, and expert usefulness ratings.
- Scaling evaluations on 200–1000 vignettes to improve subgroup and interaction statistical power.
- Extension of Proposition 1 to non-linear (neural) stale-context projection.
- Multi-center deployment studies with live EHR integration and real clinician participants.

6 Conclusion

Geodesic Diagnostic Trajectories (GDT) introduces a differential-geometry framework for adaptive clinical information retrieval that formally bounds stale-context interference, detects abrupt diagnostic pivots via geodesic curvature, and applies confidence-scaled optimal-control prefetching to anticipate clinician information needs. In a randomized four-arm crossover benchmark across 124 sessions from 31 clinical vignettes, GDT achieved the largest latency reduction among all conditions (26.9% vs. TF-IDF, $p < 0.001$, $d = -4.12$; 4.0% additional gain over CMA) and the largest simulated cognitive load reduction ($d = -0.55$). Proposition 1 was empirically verified across 100% of gate-triggered turns, establishing GDT’s theoretical guarantees on a clinical multi-pivot benchmark.

A critical finding is that all four conditions—including GDT and the stronger BM25 baseline—achieved 0% task completion, confirming that sparse lexical retrieval is insufficient for multi-pivot, multi-domain clinical chart-review regardless of session-management strategy. GDT’s geodesic-shift detection can reliably identify pivot points (34.7% trigger rate), but translating this detection into retrieval-quality improvements requires a dense retrieval backend with clinical fine-tuning.

The M/M/c operational model projects a 4.1-minute reduction in mean patient processing time per reviewer during peak-acuity periods, connecting sub-second retrieval improvements to organizationally meaningful outcomes. GDT’s formal theoretical foundations, together with its demonstrated latency and cognitive-load gains, establish a rigorous basis for future human-in-the-loop studies integrating dense retrieval, federated learning across clinical sites, and live EHR deployment.

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